LLM Fine-Tuning

With Quantized Low-Rank Adaptation

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Agenda

1. Attention Mechanism
2. Fine-Tuning
3. Low-Rank Adaptation
4. Hands-on workshop

Attention Mechanism

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Conceptualizing Attention

The “Soft” Lookup

Standard databases do "Hard Lookups" (ID=105 gets exactly ID=105).

Attention is probabilistic:

- **Query:** "Feline pets"
- **Key:** Tags like "Cat", "Dog"
- **Result:** 90% "Cat", 10% "Dog"

The Mechanism

We combine elements via a **Weighted Sum**.

1. **Score:** Dot Product of Q and K.
2. **Weight:** Softmax to sum to 1.0.
3. **Extract:** Multiply weights by Value (V).

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Multi-Head Architecture

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}V\right)$$

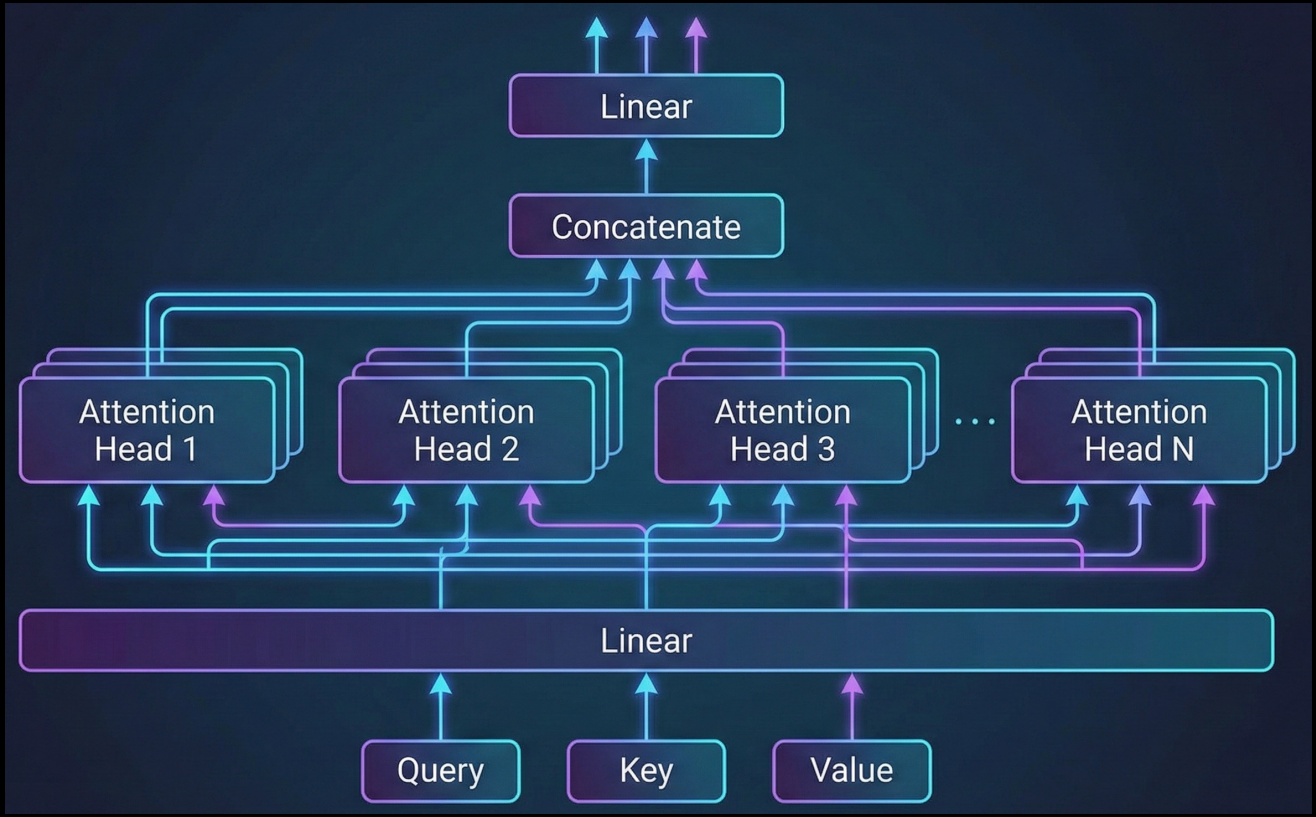
Why Multi-Head

A single head focuses on one thing.

Head 1: Syntax (Subject-Verb)

Head 2: Semantics (Pronouns)

We compute **h** heads in parallel and concatenate.



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Fine-Tuning

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Fine-Tuning Strategy

The “Last Mile”

Foundation models are generalists. Fine-tuning specializes them for your internal schema, jargon, or coding style.

Bridges general capabilities and specific utility

Vs. RAG

- **RAG:** External facts (Stock prices).
- **Fine-Tuning:** Deep behavior (Tone, JSON formatting, Safety).

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Data Requirements

Supervised (SFT)

Requires **Instruction-Response** pairs.

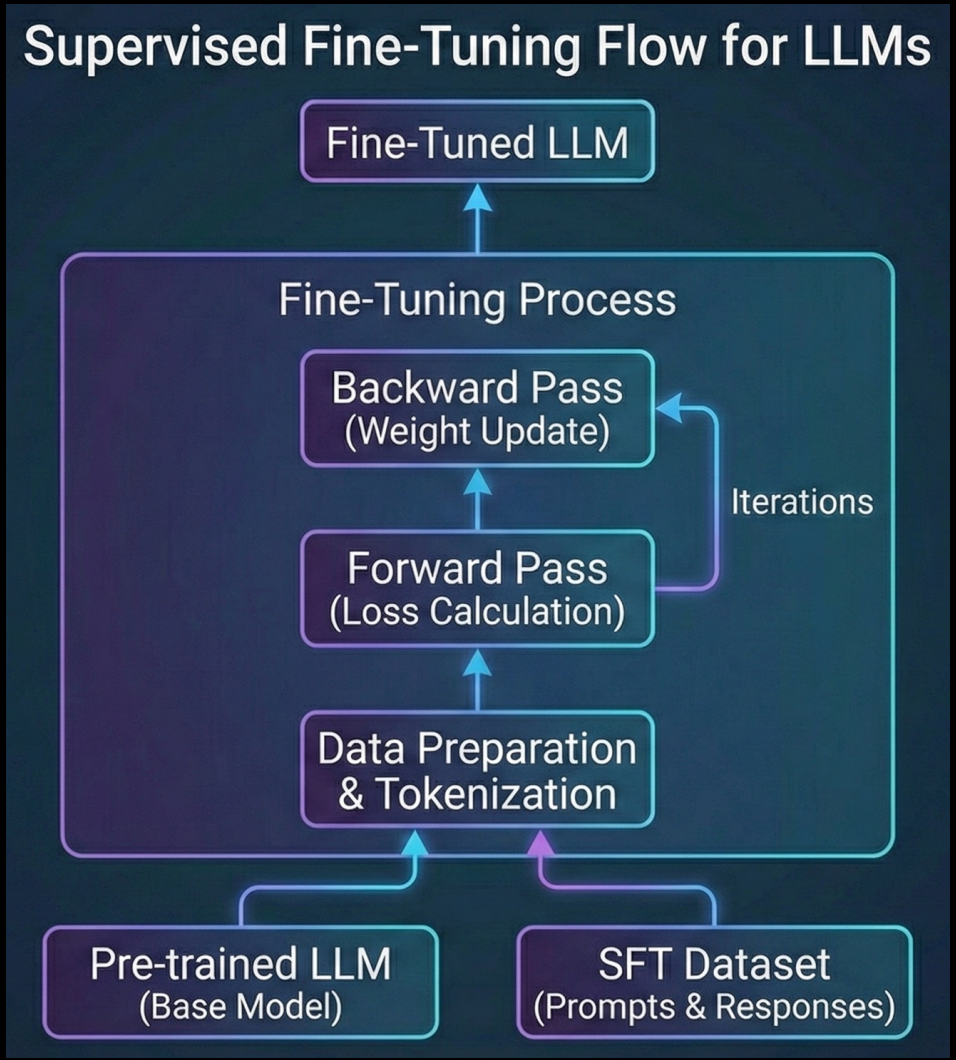
Lima Hypothesis

Quality > Quantity. **1,000 clean examples** often beat 50k noisy ones.

Formatting

```
<|user|> Prompt <|end|>
```

```
<|model|> Response <|end|>
```



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Knowledge Distillation

Teacher - Student

Compress a massive Teacher (70B) into a fast Student (7B).

Soft Labels

The Student learns the Teacher's **probability distribution** (uncertainty), not just the final answer.

Use Case

Edge devices, robotics, and low-latency APIs where big models are too slow.

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Low-Rank Adaptation

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LoRA: Low-Rank Adaptation

Weight updates ΔW have low intrinsic rank.

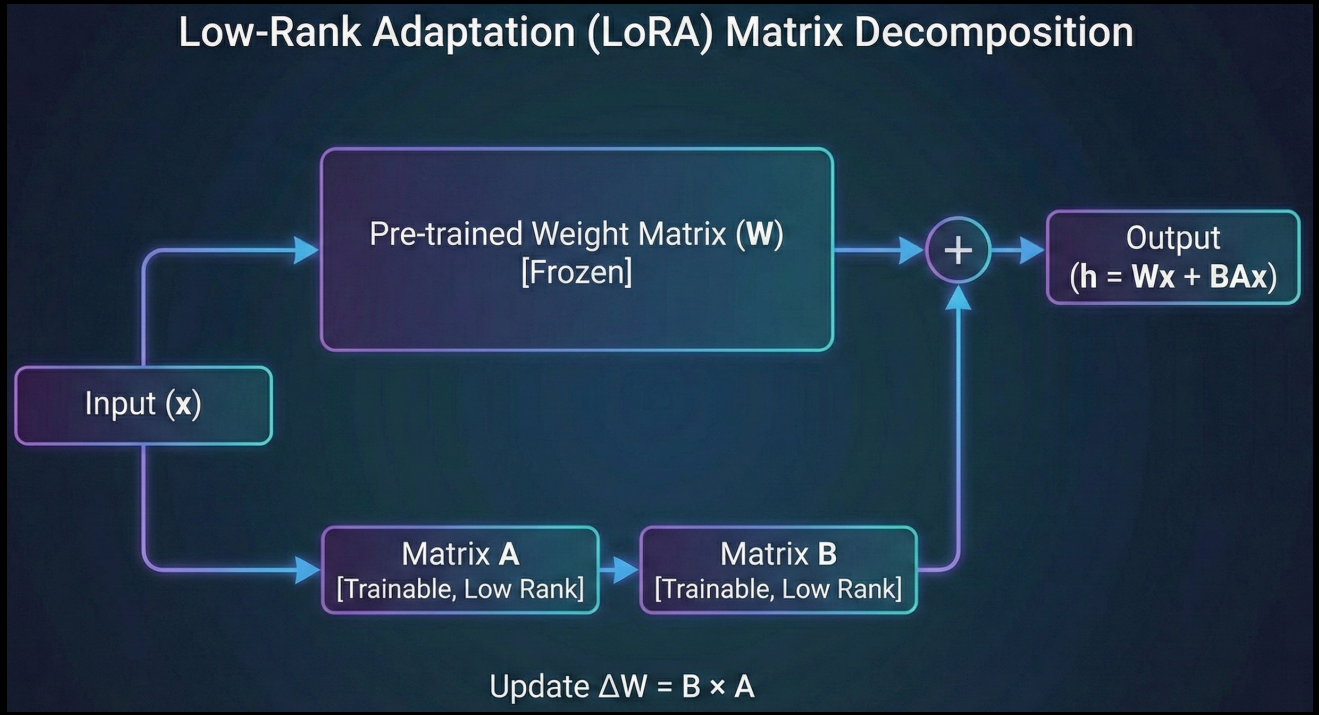
Mechanism

Freeze huge matrix W. Train tiny matrices A and B.

$$W + \Delta W = W + BA$$

Reduces trainable parameters by 10,000x.

QLoRA applies this to attention and feed-forward layers.



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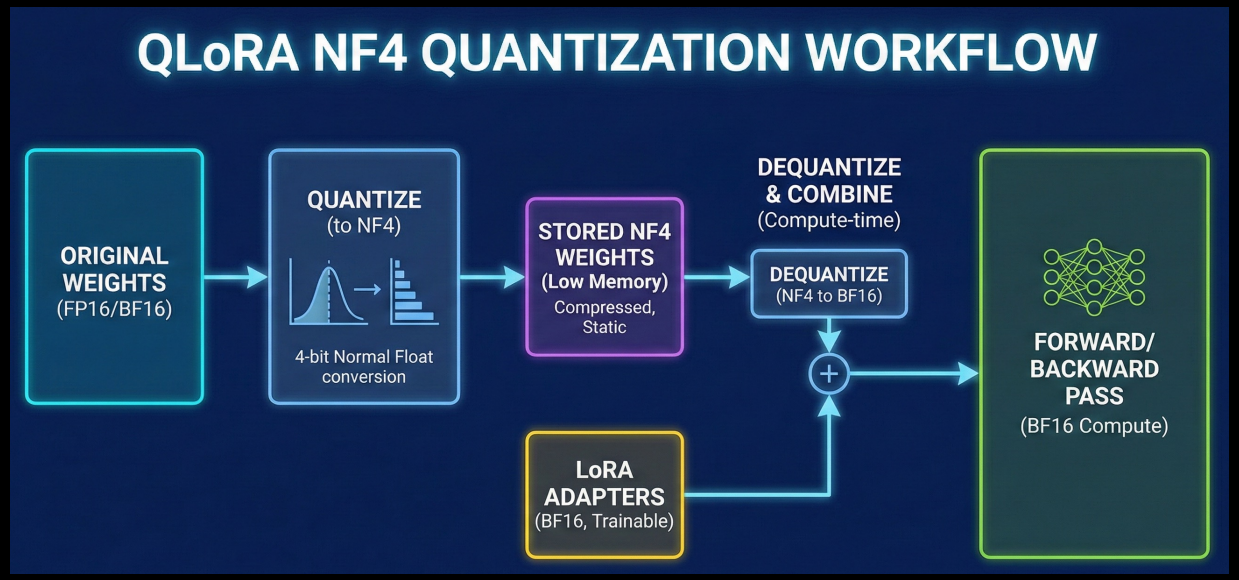
QLoRA: Quantization + LoRA

The Problem

Base model W consumes massive VRAM.

Innovations

1. **4-bit NormalFloat:** Model weights are quantized to 4-bit, generating a list of FP32 scale factors (one per block). 1bit sign, 3bits lookup to B16
2. **Double Quantization:** Scale factors are quantized as well (again in blocks)
3. **Paged Optimizers:** CPU offloading to prevent OOM.



After training

$$W_{merged} = W_{base} + \Delta W = W_{base} + (B \times A \times scaling)$$

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Evaluation: ROUGE

The Metric

Precision-Recall oriented n-gram overlap. Standard for summarization.

Checks: "Did we capture the key n-grams?"

Calculation ROUGE-2

Ref: "the cat was under the table"

Cand: "the cat was found under the table"

Matches: {the cat, cat was, under the, the table}

$$Recall = \frac{Matches}{Ref_N} = \frac{4}{5} = 0.80$$

$$Precision = \frac{Matches}{Cand_N} = \frac{4}{6} = 0.67$$

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Hands-On

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AWS Workshop

- Access link:
- Github repo: https://github.com/aws-samples/generative-ai-on-amazon-sagemaker/tree/main/workshops/fine-tuning-with-sagemakerai-and-bedrock
- Recommended: Task 1, Task 2, Task 3, Task 4
- Not required: Task 5, Distillation Task
- **Note: No need to enable FSDP - config is commented out**
- Distillation task: https://github.com/SaricVr/knowledge-distillation-transformers-pytorch-sagemaker - sagemaker-distillation.ipynb

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Distillation task

- Repo: https://github.com/philschmid/knowledge-distillation-transformers-pytorch-sagemaker/tree/master - sagemaker-distillation.ipynb

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Thank you!

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